```
def player_turn(board, player):
    while True:
        try:
                row = int(input(f"Player {player}, enter the row (0-2): "))
                col = int(input(f"Player {player}, enter the column (0-2): "))
                if board[row][col] == ' ':
                    board[row][col] = player
                    break
                else:
                    print("This cell is already occupied. Try again.")
        except (ValueError, IndexError):
            print("Invalid input. Please enter row and column values between 0 and 2.")
```

Explanation:

- The `player_turn()` function takes the board and the current player ('X' or 'O') as arguments.
- It prompts the player to input a row and column index for their move.
- If the chosen cell is empty (`''`), the player's mark is placed there. Otherwise, the player is asked to try again.
- The `try-except` block ensures that invalid input (non-integer values or out-of-bounds indexes) is handled gracefully.
